EAS 4870
HW 1

1 $\underline{E}_1 = E_0 (2j\hat{x} + \hat{y}) e^{j(\omega t - kz)}$

$\underline{E}_2 = E_0 (\hat{x} + e^{j\pi/4}\hat{y}) e^{j(\omega t - kz)}$

$|\underline{E}_1|^2 = \underline{E}_1 \cdot \underline{E}_1^* = E_0^2 (2j\hat{x} + \hat{y}) \cdot (-2j\hat{x} + \hat{y}) = 5E_0^2$

$|\underline{E}_2|^2 = E_0^2 (\hat{x} + e^{j\pi/4}\hat{y}) \cdot (\hat{x} + e^{-j\pi/4}\hat{y}) = 2E_0^2$

$\underline{E}_1 \cdot \underline{E}_2^* = E_0^2 (2j\hat{x} + \hat{y})(\hat{x} + e^{-j\pi/4}\hat{y}) = E_0^2 (2j + \underbrace{e^{-j\pi/4}}_{\frac{1}{\sqrt{2}}(1-j)})$

$\langle P \rangle = \frac{1}{2Z_0} |\underline{E}_2|^2 = \frac{E_0^2}{Z_0}$ $\qquad = E_0^2 \left(\frac{1}{\sqrt{2}} + \left(2 - \frac{1}{\sqrt{2}}\right)j\right)$

$\underline{E}_2\Big|_{z=0} = E_0 (\hat{x} + \underbrace{e^{j\pi/4}}_{\frac{1}{\sqrt{2}}(1+j)}\hat{y}) [\cos\omega t + j\sin\omega t]$

REAL PART: $E_0 \underbrace{\cos\omega t}_{\phi} \hat{x} + \frac{E_0}{\sqrt{2}} (\cos\omega t + \sin\omega t)\hat{y}$ SEE PARAMETRIC PLOT (APPENDIX)

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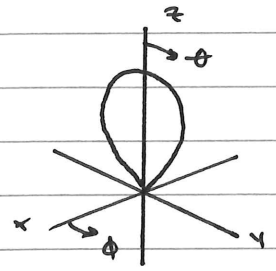

$P_r \propto \cos^n\theta$ (UPPER HALF SPACE)

$$D = \frac{P_r}{\langle P_r \rangle} = \frac{P_r}{\int P_r \, d\Omega \, \frac{1}{4\pi}} = \frac{4\pi \cos^n\theta}{2\pi \int_0^{\pi/2} \cos^n\theta \sin\theta \, d\theta \, d\phi}$$

$$= \frac{2\cos^n\theta}{\left.\frac{\cos^{n+1}\theta}{n+1}\right|_{\pi/2}^{0}} = \frac{2\cos^n\theta}{(n+1)^{-1}} \xrightarrow{\text{MAX}} 2(n+1)$$

NEED $n = 9$ FOR $D = 20$

3. $$F(\theta) = \left(\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right)^2$$ FOR DIPOLE $\parallel \hat{z}$

SAY DIPOLE IN $\hat{d}$ DIRECTION

NOTE $\cos\theta = \hat{d}\cdot\hat{r}$

$\sin^2\theta = 1 - \cos^2\theta$

SO $$F(\theta,\phi) = \frac{\cos^2\left(\frac{\pi}{2}\hat{d}\cdot\hat{r}\right)}{1-\cos^2\theta} = \frac{\cos^2\left(\frac{\pi}{2}\hat{d}\cdot\hat{r}\right)}{1-(\hat{d}\cdot\hat{r})^2}$$ GENERAL

IF $\hat{d} \parallel \hat{x}$, $\hat{d}\cdot\hat{r} = \sin\theta\cos\phi$

$\hat{d} \parallel \hat{y}$, $\hat{d}\cdot\hat{r} = \sin\theta\sin\phi$

SUBSTITUTE ABOVE

4 Integrate $\int_0^{\pi} \left(\frac{\cos(\pi/2 \cos\theta)}{\sin\theta} \right)^2 \sin\theta \, d\theta$

$\rightarrow$ 1.219 (appendix)

5. in polar coordinates, $\int_0^{\pi/2} \int_0^1 e^{-r^2} r \, dr \, d\theta = \frac{\pi}{4} e^{-r^2} \Big|_1^0 = 0.496$